

Necessity Entrepreneurship*

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Abstract

This paper studies when higher entrepreneurship reflects productive opportunity rather than weak labor-market outside options. We develop a model of occupational choice with search frictions, unemployment insurance, and two entrepreneurial margins: nonemployer self-employment and employer entrepreneurship. Necessity entrepreneurs choose business ownership only when wage-work options are weak, whereas opportunity entrepreneurs would choose entrepreneurship even with strong outside options. Quantitatively, lower unemployment insurance raises entrepreneurship mainly through low-scale self-employment, while stronger labor-market security shifts entrepreneurial activity toward employer firms and increases aggregate output per household. The main lesson is that the same rise in business ownership can imply very different aggregate outcomes depending on whether it reflects improved entrepreneurial opportunities or deteriorating labor-market insurance.

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1 Introduction

Entrepreneurship is often treated as a marker of innovation, firm growth, and rising wages. But entrepreneurial entry can reflect very different underlying choices: some individuals choose it despite attractive wage-work options, while others do so only when those options are weak or unavailable. That distinction matters for aggregate efficiency. If business ownership rises mainly because individuals have poor outside options, then higher entrepreneurship may reflect weak labor-market opportunities rather than productive entrepreneurial dynamism. In this paper, we formally distinguish between opportunity and necessity entrepreneurship within an occupational-choice framework. Opportunity entrepreneurs are individuals who choose entrepreneurship even when viable wage employment is available, whereas necessity entrepreneurs choose entrepreneurship only when it is not.

This distinction is likely to vary systematically across individuals. The same characteristics that make an individual more productive as an entrepreneur may also improve their prospects in wage work. If higher human capital raises both managerial potential and labor-market opportunities, then entrepreneurship must clear a higher bar to be optimal. By contrast, weaker outside options may push lower-human-capital individuals into entrepreneurial activity under less favorable conditions. As a result, changes in entrepreneurial entry can have very different aggregate implications depending on who enters and what type of business activity they undertake.

The difficulty is that this distinction is not directly observed. Empirical work has therefore relied on indirect measures, but these do not cleanly uncover the underlying occupational-choice object. One approach uses stated motives. In weighted U.S. GEM APS microdata for 2019–2021, between 41 and 50 percent of early-stage entrepreneurs report starting a business because jobs are scarce, while between 65 and 73 percent also report motives tied to building wealth or very high income. These responses are informative, but they do not deliver a clean classification: necessity and opportunity signals can coexist within the same entrepreneurial spell. A second approach classifies entrepreneurs by prior labor-market status. That is also useful, but observed status at entry is not the object we seek to identify. Workers may start a business while still employed because they anticipate worsening job prospects, while some more growth-oriented entrepreneurs may pass through temporary non-employment before starting a firm.

What is missing, however, is a definition of this distinction at the level of occupational choice. In the model below, entrepreneurship can take the form of nonemployer self-employment or employer entrepreneurship. This allows the framework to speak not only to entrepreneurial entry, but also to the composition and scale of entrepreneurial activity. In particular, it can distinguish between an increase in low-scale business ownership driven by weak outside options and an increase in employer-oriented entrepreneurship driven by stronger entrepreneurial prospects.

The paper makes three contributions. First, it offers a formal definition of necessity entrepreneurship grounded in optimal occupational choice rather than stated motives or prior labor-market status. Second, it develops a theoretical framework that embeds search and match-

ing frictions within a span-of-control model of occupational choice, so the wage-work outside option is endogenous to labor-market conditions. Third, the quantitative results show that policy-induced increases in entrepreneurship operate mainly through a compositional margin: weaker outside options raise low-scale self-employment much more than employer entrepreneurship. This is why the paper emphasizes entrepreneurial composition, scale, and aggregate output rather than the entrepreneurship rate alone.

The remainder of the paper is organized as follows. Section 2 presents the empirical motivation. Section 3 discusses the related literature. Section 4 sets out the model environment. Section 5 characterizes equilibrium. Section 6 discusses calibration. Section 7 presents the quantitative results. Section 8 studies policy counterfactuals. Section 9 concludes.

2 Empirical Motivation

Necessity entrepreneurship is difficult to identify cleanly in observed data. But recessions are informative because they weaken outside options in wage employment.

Figure 1 shows this pattern in business stocks. During the Great Recession, zero-employee establishments rose by 1.85 percent, while businesses with employees contracted. The decline was largest among establishments with 20 or more employees, which fell by 5.82 percent. Lower-scale business ownership was therefore comparatively resilient even as employer firms weakened. This does not mean that nonemployer businesses directly measure necessity entrepreneurship. It does suggest that downturns shift entrepreneurial activity toward smaller-scale ownership rather than employer growth.

Figure 1: U.S. business counts by employment size, indexed to 1997.

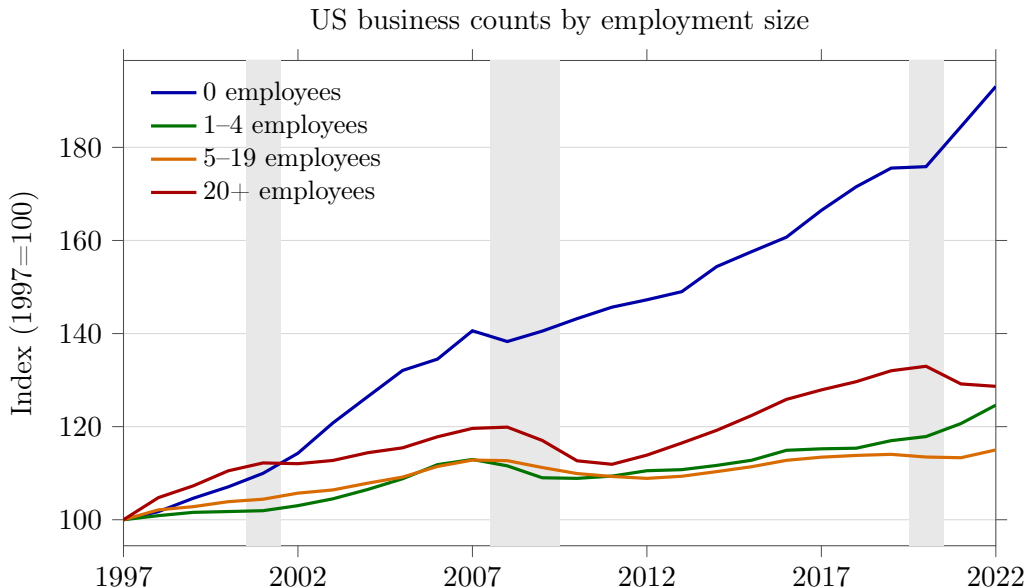
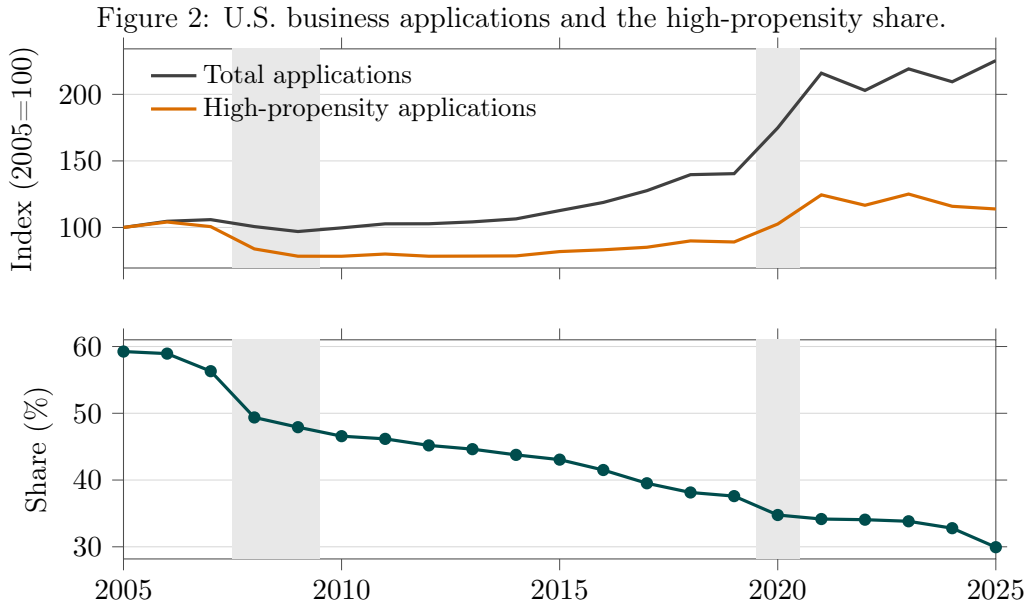


Figure 2 shows the same pattern in business formation flows. Total business applications fell by 5.84 percent during the recession, but high-propensity applications fell by 22.12 percent, and their share of total applications also declined sharply. The recession therefore did not simply reduce entry in the aggregate; it shifted new business formation away from the employer-oriented margin.



Taken together, the figures motivate the paper’s main measurement point: different empirical measures capture different margins of entrepreneurship, and these should not be treated as economically equivalent. The framework developed below is designed to explain these compositional differences rather than a single aggregate entrepreneurship statistic.

3 Literature Review

This paper relates most directly to the literature distinguishing opportunity from necessity entrepreneurship. Much of the empirical literature operationalizes this distinction using observable proxies, especially stated motives or pre-entry labor-market status. Fairlie and Fossen (2019) propose an objective classification based on pre-entry work status and show that opportunity and necessity entry display different cyclical patterns. Fairlie (2013) and Fossen (2021) likewise show that entrepreneurial entry is heterogeneous over the cycle, with unemployment-driven and lower-scale entry behaving differently from more growth-oriented business creation. Using GEM microdata, Chivers (2017) show that necessity entrepreneurship is negatively associated with economic development, whereas opportunity entrepreneurship is positively associated with development. Poschke (2013) provide the closest bridge from these empirical patterns to a structural interpretation by emphasizing how labor-market prospects shape entrepreneurial

selection, and Levine and Rubinstein (2020) further show that entrepreneurship and other forms of self-employment should not be treated as interchangeable objects. This literature makes clear that entrepreneurial entry is heterogeneous, but it usually measures that heterogeneity through reduced-form proxies rather than through the occupational-choice problem itself.

The paper also builds on the occupational-choice tradition in macroeconomics and entrepreneurship. Lucas (1978), Evans and Jovanovic (1989), and Buera (2009) show why entrepreneurial selection, firm scale, and financial constraints should be studied jointly rather than through entry alone. Hurst and Pugsley (2011) add an important empirical qualification: much small-business activity is not well described by the standard high-growth view of entrepreneurship. That point is central here. If weaker outside options raise primarily low-scale self-employment rather than employer entrepreneurship, then the relevant outcome is not simply whether business ownership rises, but which type of business activity expands and with what implications for firm scale, output, and welfare.

The paper also relates to a smaller but growing literature that embeds self-employment in equilibrium search environments. Mortensen and Pissarides (1994) and Petrongolo and Pissarides (2001) provide the benchmark search-and-matching framework for job-finding rates, separations, and labor-market tightness. Albrecht et al. (2009) and Satchi and Temple (2009) show that in economies with formal and informal sectors, search frictions shape the allocation of workers across unemployment, wage work, and self-employment. Narita (2020), Denderski and Sniekers (2024), and Poschke (2025) move closer to the present paper by modeling self-employment as an alternative to job search and showing that changes in labor-market frictions materially affect self-employment outcomes. Chodorow-Reich and Karabarbounis (2016) add that the opportunity cost of employment is itself cyclical, while Hombert et al. (2020) show that unemployment insurance can raise entrepreneurial entry by reducing downside risk. The present paper brings these strands together in a setting where job-finding risk, separation risk, and unemployment insurance jointly shift the occupational-choice threshold, making the distinction between opportunity and necessity entrepreneurship endogenous to labor-market conditions.

The paper also relates to the literature on entrepreneurial composition over the business cycle. Sedlacek and Sterk (2017) show that startup employment is procyclical and that cohort differences reflect startup size and growth potential, not only startup counts. Haltiwanger et al. (2013) and Fort et al. (2013) similarly show that young firms, startup job creation, and firm scale respond strongly to cyclical conditions. This evidence suggests that changes in entrepreneurship should not be interpreted through a single extensive-margin statistic. That perspective aligns closely with the central prediction of this paper: weaker outside options can increase lower-scale business ownership even when employer-oriented entry or startup growth potential weakens.

Relative to these literatures, the paper contributes in two ways. First, it offers a structural definition of necessity entrepreneurship grounded in optimal occupational choice rather than in motives or prior status alone. Second, it embeds that distinction in a framework with heterogeneous

human capital, entrepreneurial ability, search frictions, closure risk, and social insurance, allowing the model to speak not only to entrepreneurial entry, but also to entrepreneurial composition, firm scale, and the distinction between self-employment and employer entrepreneurship.

4 The Model: Economic Environment

Consider a Lucas (1978) span of control model, where individuals differ in the level of human capital, h . As in Gomez and Kuhen (2017), individuals are educated to either the primary (P), secondary (S) or tertiary level (T). The level of education means that a tertiary educated individuals have a higher draw of human capital, on average, than secondary educated individuals, who in turn have a higher draw on average than primary educated individuals. This human capital draw is positively correlated with managerial ability x as well as labor productivity z .

At the start of the period agents enter as either unemployed ($i_w = 0$), employed ($i_w = 1$), or an entrepreneur ($i_w = 2$). Individuals face a choice of whether to become an entrepreneur, to continue working or to search for work. However, the value of employment is different depending on whether an agent enters the period as employed ($i_w = 1$) or not. Although there is a risk that an employee may become separated from a firm with probability ρ , this is less risky than searching for work which has a $p(\theta)$ probability of obtaining employment. Consequently, the expected value of employment is higher for those already in employment than those who are not. Additionally, the probability of separation is lower for individuals with higher levels of human capital and they are able to command a higher wage due to higher levels of productivity. So the value of employment, holding other things equal, is higher for those with higher human capital as well.

The innovation of this model is that the entrepreneurial decision results in two types of entrepreneurs emerge, opportunity-based entrepreneurs and necessity-based entrepreneurs. We define opportunity-based entrepreneurs are those whose value of entrepreneurship is higher, even if they had employment. Necessity-based entrepreneurs are those who would not have become entrepreneurs, if they had employment.

Timing: The timing of events can be summarized below. Occupational choice happens at the start of time t . Those choosing to become entrepreneurs run projects and choose capital and hired labor each period. If hired labor satisfies $n_h = 0$ the business is nonemployer self-employment, while $n_h > 0$ corresponds to employer entrepreneurship. Entrepreneurial businesses face closure risk $\rho_e(h)$. In the benchmark, closure moves the household out of business and into non-employment without automatic regular unemployment insurance eligibility. Those who were previously employed within a project can decide to carry on working but carry a risk of separation with probability ρ . If separation occurs these workers will claim unemployed benefits and be unable to look for work until the following period. After separation occurs, vacancies are posted and who were unemployed in the previous period are matched with firms. Those that are

successful and match with with probability $p(\theta)$ go on to earn a wage. Whereas those that are unsuccessful will claim unemployment benefits and be unable to search again until the following period.

Unlike the search and matching literature there is no wage bargaining between workers and the firm at the end of matching. Instead the wage pins down the occupational choice decision which determines those wanting to become entrepreneurs and those who will search for work. Crucially, the worker only finds out whether they enter production and earn a wage after their occupational choice decision.

Figure 3: Timing for Matched Workers

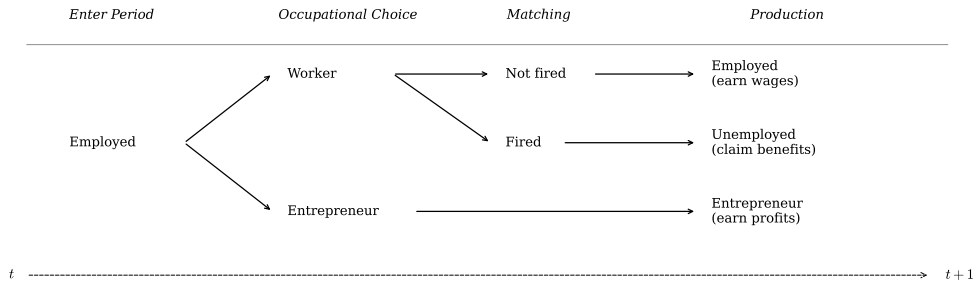
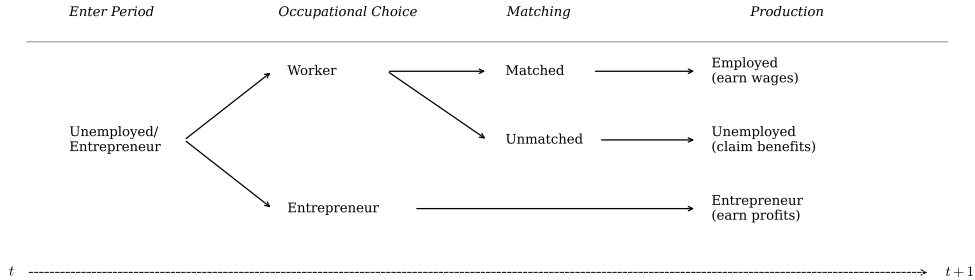


Figure 4: Timing for Unmatched Workers



4.1 Preferences, endowments and technology

Preferences: Consumption by an agent in period t is c_t , with utility given by $U(c_t)$.

Endowments: Agents are endowed with an initial capital asset, a_0 , which can be used as an input in production. Each individual is endowed with human capital shock h^i . Human capital h is positively correlated with managerial talent x , labor productivity z and separation rate ρ .

Production: Firms use labor n and capital k to produce a single consumption good, y . Capital depreciates at a constant rate of δ . Managers can operate only one project. The functional form of the production function is:

$$y = xk^\alpha (\ell_e + \xi zn_h)^\gamma \quad \text{where } \alpha, \gamma > 0. \quad (1)$$

so self-employment is feasible at $n_h = 0$, while employer entrepreneurship corresponds to $n_h > 0$. Managerial talent x is determined by the Pareto distribution $F(x) = \left[1 + \frac{x-\varsigma}{\xi}\right]^{-\vartheta}$ where ϑ is the shape parameter, ς the location parameter and ξ is the scale parameter.

Factor remuneration: Firms rent capital at the common market rate r .

Government: The government runs a balanced budget each period and provides unemployment insurance which is financed through lump sum taxation τ , which is shared by the employee $(1 - \psi)\tau$ and employer $\psi\tau$. This program guarantees each household an unemployment benefit of b .

4.2 The Firm's problem

The firm's problem is:

$$\max_{n,k} Xk^\alpha n^\gamma - (1 + \tau\psi)wzn - rk - \kappa \frac{\rho zn}{q(\theta_t)} \quad (2)$$

employees must pay a share of unemployment insurance τ to all workers who produce output. Firms have to pay search costs κ for all vacancies posted. The number of vacancies posted is designed to replace any lost workers, ρzn . This, however, is affected by the firm-hiring rate $q(\theta_t) = \bar{m} \left(\frac{v_t}{u_t}\right)^{-\mu}$, where aggregate vacancies v_t is simply the aggregate level of all those who have separate from firms $\sum \rho zn$ within the period and u_t is the total unemployed looking for work within that period (see appendix).

When the market is tight and the hiring rate is low, firms have to search more and thus face higher search costs in order to replace lost workers. This is because $0 < q(\theta_t) < 1$, so a lower $q(\theta_t)$ increases $\frac{\rho zn}{q(\theta_t)}$. If the firm lose 10 workers, they want to hire 10 workers to replace them. But if the hiring rate, $q(\theta_t)$, is 50% then they will face double the amount of search costs than if $q(\theta_t)$ were 100%. The upshot of this is that when markets are tight, the size of the desired firm size n decreases ex ante, as the cost of replacing workers is high and so entrepreneurs would prefer to run smaller firms.

To avoid issues of directed search, we assume that employees are randomly matched with firms (as in Chivers et al., 2015) and each firm employees a distribution of workers across the human capital spectrum such that each firm employs the market average.

5 Optimal behavior and equilibrium

In any time period t agents are distinguished by (h, a, x, i, z) . The timing of the economy is given as follows.

1. Households enter each new period with assets a and an employment status i_w
2. Idiosyncratic shocks h is drawn which determines x (managerial ability) and z (labor productivity), ρ separation rate conditional on education level, P, S or T .
3. Households make an occupation decision: entrepreneur ($\mathcal{I}_e = 1$) or worker ($\mathcal{I}_e = 0$).
4. Capital and labor markets clear. A fraction ρ of employees separate from firms and $p(\theta)$ are hired to replace them
5. Households choose: consumption (c), borrowing/saving (a')

5.1 Firm manager

Firms are distinguished by their productivity realization x . In the self-employment version, entrepreneurs choose capital and hired labor while also supplying owner labor to the business. Let n_h denote hired labor and ℓ_e denote owner labor. When $n_h = 0$ the entrepreneur is self-employed; when $n_h > 0$ the entrepreneur is an employer. The static problem for a manager with talent x^i is

$$\max_{k, n_h \geq 0} xk^\alpha (\ell_e + \xi z n_h)^\gamma - zw(1 + \psi\tau)n_h - rk - \kappa \frac{\rho z n_h}{q(\theta)} - f_{hire} \mathbf{1}\{n_h > 0\} \quad (3)$$

Self-employment is therefore the corner $n_h = 0$, while employer entrepreneurship corresponds to $n_h > 0$. For an interior employer choice, the first-order condition for hired labor is:

$$\gamma x k^\alpha (\ell_e + \xi z n_h)^{\gamma-1} \xi z = w(1 + \psi\tau) + \frac{\rho \kappa}{q(\theta)} \quad (4)$$

$$n_h^*(k, x, \tilde{w}) = \max \left\{ 0, \frac{1}{\xi z} \left[\left(\frac{\gamma x k^\alpha \xi z}{w(1 + \psi\tau) + \frac{\rho \kappa}{q(\theta)}} \right)^{\frac{1}{1-\gamma}} - \ell_e \right] \right\} \quad (5)$$

Substituting (5) into (3) yields the entrepreneur's profit function:

$$\pi(k, n_h^*, x; \tilde{w}) = xk^\alpha (\ell_e + \xi z n_h^*)^\gamma - z n_h^* w(1 + \psi\tau) - rk - \kappa \frac{\rho z n_h^*}{q(\theta)} - f_{hire} \mathbf{1}\{n_h^* > 0\} \quad (6)$$

5.1.1 Capital

Initially we assume capital markets function perfectly, so the interior capital choice satisfies

$$\alpha x k^{\alpha-1} (\ell_e + \xi z n_h)^\gamma = r.$$

5.2 Entrepreneurs

Entrepreneurs maximize expected discounted utility of consumption

with $n_h = 0$ denoting nonemployer self-employment and $n_h > 0$ denoting employer entrepreneurship.

$$\max_{\{c_t, a_{t+1}, k_t, n_{h,t}\}} \mathbb{E} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + \pi(k, n_h, x; \tilde{r}, w, \tau) - \tau_\pi \quad (7)$$

$$a' \geq -\bar{a}$$

5.3 Employed

The workforce (previous periods employed who choose to reapply for work) maximize expected discounted utility of consumption

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + (1 - \rho)(wz(1 - (1 - \psi)\tau_y)) + \rho b \quad (8)$$

$$a' \geq -\bar{a}$$

where b is the unemployment benefit.

5.4 Unemployed

The unemployed maximize expected discounted utility of consumption

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + p(\theta)(wz(1 - (1 - \psi)\tau_y)) + (1 - p(\theta))b \quad (9)$$

$$a' \geq -\bar{a}$$

where b is the unemployment benefit.

5.5 Government

The government runs a balanced budget with a lump-sum taxes $T(y) = (y - \lambda_p y^{1-\tau_p})$

5.6 The household's problem

Let i_w denote the household's labor market status in the previous period, with $i_w = 2$ indicating being entrepreneur, $i_w = 1$ being employed and $i_w = 0$ being unemployed. Let $\mathcal{I}_e$ indicate occupational choice, where $\mathcal{I}_e = 1$ if the household is an entrepreneur and $\mathcal{I}_e = 0$ if the household is a worker.

Let $\mathbb{E}_{\rho(h)}$ denote the continuation operator for employed workers who face job separation, let $\mathbb{E}_{p(\theta)}$ denote the continuation operator for households searching for work, and let $\mathbb{E}_{\rho_e(h)}$ denote the continuation operator for entrepreneurs facing business closure. We write $\mathbf{V}_0$ for the continuation value after business closure; in the benchmark this state does not automatically receive regular unemployment insurance. More specifically,

$$\mathbb{E}_{p(\theta)} \mathbf{V}(a', h, x', i'_w) = p(\theta) \mathbf{V}(a', h, x', i'_w = 1) + (1 - p(\theta)) \mathbf{V}(a', h, x', i'_w = 0)$$

$$\mathbb{E}_{\rho(h)} \mathbf{V}(a', h, x', i'_w) = (1 - \rho(h)) \mathbf{V}(a', h, x', i'_w = 1) + \rho(h) \mathbf{V}(a', h, x', i'_w = 0)$$

$$\mathbb{E}_{\rho_e(h)} \mathbf{V}(a', h, x', i'_w) = (1 - \rho_e(h)) \mathbf{V}(a', h, x', i'_w = 2) + \rho_e(h) \mathbf{V}_0(a', h, x', i'_w = 0)$$

$$\mathbf{V}(a, h, x, i_w) = \max \{V^w(a, h, x, i_w), V^e(a, h, x)\}$$

$$V^w(a, h, x, i_w) = \max_{\{a', c\}} \{u(c) + \iota_{i_w=1} \beta \mathbb{E}_{\rho(h)} \mathbf{V}(a', h, x', i'_w) + \iota_{i_w \neq 1} \beta \mathbb{E}_{p(\theta)} \mathbf{V}(a', h, x', i'_w)\}$$

$$V^e(a, h, x) = \max_{\{a', c, k, n_h \geq 0\}} \{u(c) + \chi_{se} \mathbf{1}\{n_h = 0\} + \chi_{emp} \mathbf{1}\{n_h > 0\} + \beta \mathbb{E}_{\rho_e(h)} \mathbf{V}(a', h, x', i'_w)\}$$

subject to

$$c + a' \leq (1 - \tilde{r} - \delta)a + inc \tag{10}$$

$$inc = \begin{cases} ra + \tilde{w}_n & \text{if } \mathcal{I}_e = 0, i_w = 1 \\ ra + \tilde{w}_u & \text{if } \mathcal{I}_e = 0, i_w \neq 1 \\ ra + \pi(k, n_h, x; \tilde{r}, w, \tau) & \text{if } \mathcal{I}_e = 1 \end{cases}$$

where inc is the earnings of the worker or entrepreneur, with entrepreneurial income coming from either self-employment ($n_h = 0$) or employer entrepreneurship ($n_h > 0$), $\tilde{w}_n = (1 - \rho) \{zw(1 - (1 - \psi)\tau)\} + \rho b$ and $\tilde{w}_u = p(\theta) \{zw(1 - (1 - \psi)\tau)\} + \{1 - p(\theta)\}b$.

5.7 Equilibrium

Because the quantitative model discretizes human capital into three education groups, the operative occupational-choice object is a status-specific managerial-ability critical value within each human-capital block. For a household with assets a , human capital h , and labor-market status $s \in \{\text{employed}, \text{unemployed}\}$, let $x_s^*(a, h)$ denote the minimum managerial ability required for entrepreneurship. Households choose entrepreneurship whenever $x \geq x_s^*(a, h)$. The value of retaining an existing wage match implies

$$x_{\text{unemp}}^*(a, h) \leq x_{\text{emp}}^*(a, h).$$

We define necessity entrepreneurship as households whose entrepreneurial ability is high enough to enter if unemployed but not high enough to enter if employed:

$$x_{\text{unemp}}^*(a, h) \leq x < x_{\text{emp}}^*(a, h).$$

We define opportunity-driven entrepreneurship as households that enter regardless of labor-market status:

$$x \geq x_{\text{emp}}^*(a, h) \geq x_{\text{unemp}}^*(a, h).$$

With three human-capital groups, the benchmark therefore yields six status-specific critical-value schedules at each asset level: employed and unemployed critical values for the low-, middle-, and high-human-capital blocks.

6 Calibration

To quantify the model, we calibrate it to match the main margins emphasized in the paper: labor-market outside options, entrepreneurial entry, and the distinction between lower-scale self-employment and employer entrepreneurship. Some parameters are disciplined externally using standard values or existing evidence, while the remaining parameters are chosen to match moments related to employment transitions, entrepreneurial entry, and firm scale. The calibration is designed not only to match aggregate entrepreneurship, but also to discipline the compositional

margins that are central to the paper. Table 1 reports the benchmark calibration.

Table 1: Benchmark self-employment calibration

Parameter	Description	Value	Source
Preferences and policy			
β	Discount factor	0.94	Benchmark model
σ	CRRRA coefficient	2.0	Benchmark model
b	UI replacement rate	0.40	Benchmark policy
$\underline{c}$	Consumption floor	0.30	Benchmark policy
a_l	Asset lower bound	0.01	Benchmark grid
Worker labor market			
$\bar{m}$	Matching efficiency	0.90	Lubik and Krause (2014)
μ	Matching elasticity	0.68	Petrongolo and Pissarides (2001)
κ	Vacancy/replacement cost	0.035	Hagedorn and Manovskii (2008)
$\rho(h)$	Separation by education	{.038, .029, .014}	Cairó and Cajner (2018)
Education and technology			
h	Population share by education	{.109, .581, .310}	Barro and Lee (2013)
z	Wage premium by education	{1.00, 1.31, 1.61}	Goldin and Katz (2008)
x -share by edu.	Span-of-control by education	{.782, .802, .822}	Benchmark calibration
Self-employment extension			
$\rho_e(h)$	Closure by education	{.168, .150, .144}	Grashuis (2021) target
λ_k	Capital lost at business closure	0.25	Benchmark choice
ℓ_o	Owner labor	1.0	Normalization
x_{SE} scale	Zero-hire productivity	1.0	Normalization
f_{hire}	Employer-threshold cost	0.25	Calibrated to match employer split

To discipline business failure, we target an average annual entrepreneur exit rate of 0.15 and use Grashuis (2021) only for the education gradient, which implies $\rho_e(h) = \{0.168, 0.150, 0.144\}$. Business closure removes the household from entrepreneurship but does not automatically grant regular unemployment-insurance eligibility. We pair that exit shock with a 25 percent loss of installed business capital. This follows the capital-illiquidity logic in Wang et al. (2012) and Chen et al. (2010), but the 25 percent loss itself is treated as a benchmark choice, so failure worsens both occupational status and next-period net worth.

The benchmark fixed employer-threshold cost is $f_{hire} = 0.25$. This cost is distinct from the proportional hiring cost governed by κ : κ captures vacancy and replacement costs that scale with hired labor, whereas f_{hire} captures the discrete hurdle between remaining self-employed and operating an employer firm. Evidence on first-worker labor costs and discrete compliance thresholds (Cockx and Desiere, 2024; Guo and Wallskog, 2025; Harju et al., 2019) motivates this wedge, but does not pin down a direct structural estimate for the benchmark value. Appendix robustness shows that the main composition results survive at nearby values of 0.15 and 0.35.

7 Benchmark Equilibrium

The benchmark makes clear that entrepreneurship is not a single object, but a composition of distinct business types and scales. Although the overall entrepreneur share is 0.178, 57.7 percent of entrepreneurs are self-employed nonemployers and 42.3 percent are employer entrepreneurs.

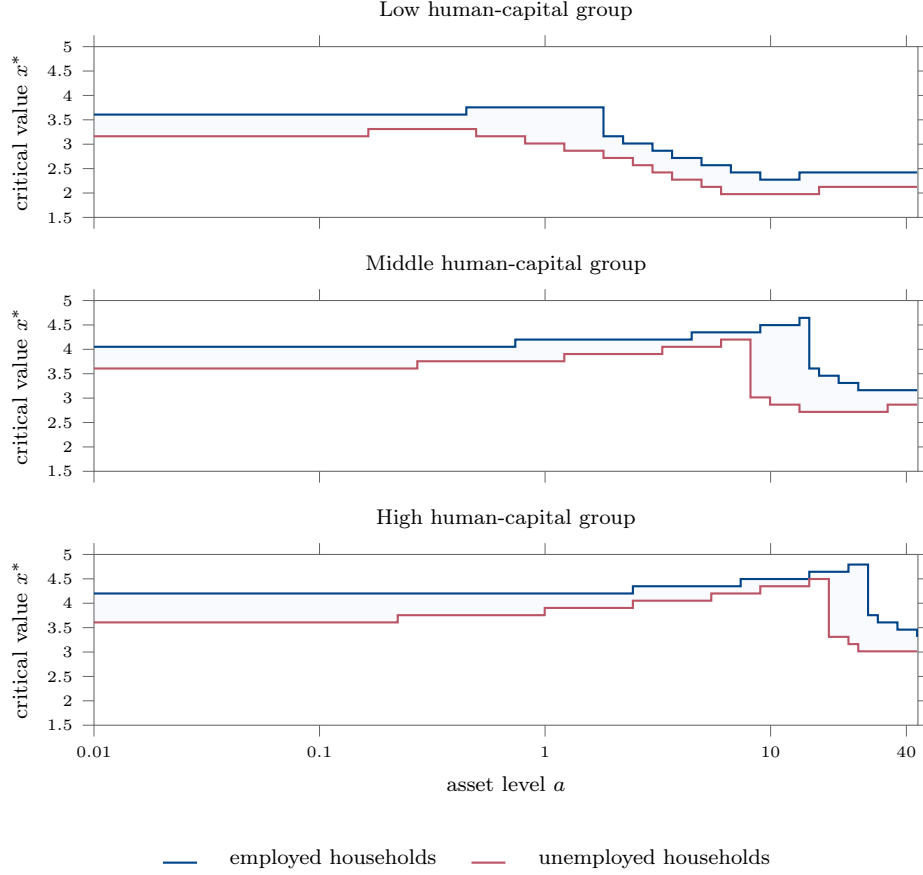
This composition varies sharply by education: lower-education entrepreneurs are concentrated in self-employment, whereas high-education entrepreneurs are much more likely to operate employer firms. Average entrepreneur labor demand is 6.16 workers. The relevant object is therefore not only the level of entrepreneurship, but also its composition by business type, education, and scale.

Table 2: Benchmark self-employment equilibrium: baseline results

	Baseline
Prices and taxes	
Interest rate r	0.097
Wage w	3.020
Job-finding rate θ	0.650
Income tax τ_y	0.007
Entrepreneur share	
All	0.178
Low-education	0.078
Mid-education	0.077
High-education	0.022
Self-employed share	
All	0.577
Low-education	0.364
Mid-education	0.199
High-education	0.015
Avg. entrepreneur size n	
All	6.159
Low-education	1.177
Mid-education	6.403
High-education	22.681
Output	
Agg. output index	1.000

Figure 5 organizes these benchmark critical values by human-capital group. Each panel fixes one human-capital block h and plots the status-specific entrepreneurial-ability critical value $x_s^*(a, h)$ over assets. Because human capital shapes both labor-market productivity and entrepreneurial returns, the location of these critical-value schedules differs across groups as well as across employment states. The unemployed schedule lies below the employed schedule in every panel, so weaker outside options lower the critical value required for entry.

Figure 5: Necessity entrepreneurship by human capital.



The shaded band identifies the necessity-entry states: households that would choose business ownership when unemployed, but not when employed, holding assets and human capital fixed. This region is narrow at very low asset levels because liquidity constraints bind in both employment states. It widens at intermediate asset levels, where households can finance entry but employed workers still place substantial value on remaining in wage work. At high asset levels, the wedge narrows again because entry becomes feasible in both states. The cutoff schedules also shift to the right with human capital, since higher human-capital groups carry higher wage productivity and lower separation risk, which raises the value of staying employed and therefore requires a higher managerial-ability critical value for entry. Read in this way, Figure 5 captures the paper's central mechanism: weaker outside options expand entrepreneurship by enlarging the necessity-entry region rather than by uniformly improving entrepreneurial opportunities.

8 Counterfactuals

We now use the calibrated model to study how labor-market conditions and policy affect both the level and composition of entrepreneurship. The exercises below are designed to isolate the paper’s central mechanism: changes in outside options do not simply raise or lower entrepreneurial entry in the aggregate, but also shift the type of entrepreneurial activity that emerges. In particular, the model distinguishes between lower-scale self-employment and employer entrepreneurship, allowing the counterfactuals to speak to composition, scale, and aggregate output.

8.1 Effect of Unemployment Insurance

The first experiment studies how unemployment insurance affects entrepreneurial entry through the wage-work outside option. In the model, lowering unemployment insurance reduces the expected value of wage employment by weakening the insurance attached to paid work. This matters even for individuals with relatively strong labor-market prospects, since employment remains exposed to separation risk. The key question is therefore not simply whether lower unemployment insurance raises entrepreneurship, but whether it does so by improving entrepreneurial opportunities or by making wage work less attractive. To study this, we reduce the unemployment-insurance replacement rate from its benchmark value of 0.40 to 0.05 and then to zero.

Table 3: Benchmark self-employment equilibrium: unemployment insurance ladder

	Benchmark	Low UI	No UI
Prices and taxes			
UI replacement rate b	0.400	0.050	0.000
Interest rate r	0.097	0.093	0.091
Wage w	3.020	3.210	3.274
Job-finding rate θ	0.650	0.650	0.615
Income tax τ_y	0.007	0.002	0.001
Entrepreneur share			
All	0.178	0.289	0.342
Low-education	0.078	0.087	0.087
Mid-education	0.077	0.177	0.223
High-education	0.022	0.025	0.032
Self-employed share			
All	0.577	0.718	0.767
Low-education	0.364	0.261	0.222
Mid-education	0.199	0.438	0.503
High-education	0.015	0.019	0.041
Avg. entrepreneur size n			
All	6.159	3.320	2.553
Low-education	1.177	0.890	0.826
Mid-education	6.403	2.554	1.883
High-education	22.681	17.469	11.902
Output			
Agg. output index	1.000	0.835	0.739

Table 3 shows that lower unemployment insurance raises entrepreneurship sharply, with the entrepreneur share increasing from 0.178 in the benchmark to 0.289 under low UI and 0.342 under

no UI. This increase, however, is concentrated overwhelmingly in self-employment rather than employer entrepreneurship. The self-employed share among entrepreneurs rises from 57.7 percent to 71.8 percent and then to 76.7 percent, while average entrepreneur size falls from 6.16 to 3.32 and then to 2.55 workers. Aggregate output per household also declines, with the output index falling from 1.000 to 0.835 and then to 0.739. Lower insurance therefore raises entrepreneurship mainly by weakening the desirability of wage work and pushing more individuals into lower-scale necessity entry, rather than by generating a broad expansion of productive employer firms.

8.2 Necessity Entrepreneurship and Labor-Market Misallocation

This experiment is designed to isolate the labor-market misallocation associated with necessity entrepreneurship. In the model, necessity entrepreneurs are individuals who choose entrepreneurship only because a viable wage-work option is unavailable. If unemployment risk is removed altogether, that channel is effectively shut down: individuals who prefer wage employment obtain it and do not lose it. The experiment therefore provides a benchmark for how occupational sorting changes when entrepreneurship no longer serves as a response to labor-market risk. To study this, we set job finding to one and worker separation to zero, while holding the benchmark tax rate fixed. This is not a fully efficient allocation, since entrepreneurial risk, borrowing limits, employer-threshold costs, and occupational heterogeneity remain in place, but it is informative about the role of labor-market frictions in generating necessity entry.

Table 4: Benchmark self-employment equilibrium: baseline versus no unemployment risk

	Benchmark	No unemployment risk
Prices and taxes		
Interest rate r	0.097	0.102
Wage w	3.020	3.091
Job-finding rate θ	0.650	0.989
Income tax τ_y	0.007	0.007
Entrepreneur share		
All	0.178	0.193
Low-education	0.078	0.050
Mid-education	0.077	0.101
High-education	0.022	0.042
Self-employed share		
All	0.577	0.397
Low-education	0.364	0.202
Mid-education	0.199	0.179
High-education	0.015	0.017
Avg. entrepreneur size n		
All	6.159	5.807
Low-education	1.177	1.539
Mid-education	6.403	4.929
High-education	22.681	13.043
Output		
Agg. output index	1.000	1.135

Table 4 shows that shutting down unemployment risk has only a modest effect on total

entrepreneurship, with the entrepreneur share rising from 0.178 to 0.193. The composition of entrepreneurship changes much more sharply. The self-employed share falls from 57.7 percent to 39.7 percent, implying a corresponding rise in the employer share from 42.3 percent to 60.3 percent. Aggregate output per household increases by 13.5 percent. The experiment therefore suggests that once the necessity channel is shut down, entrepreneurial activity is reallocated away from lower-scale self-employment and toward more productive employer firms.

8.3 Effect of a Recession Shock

The final experiment studies the effect of a recession-style deterioration in labor-market conditions. The purpose is to isolate the paper’s central mechanism in a downturn environment: when wage-work outside options weaken, some marginal individuals enter entrepreneurship, but primarily at smaller scale. Motivated by the empirical patterns in Section 2, we therefore consider a shock that worsens labor-market prospects for low-education individuals by increasing their separation rate.

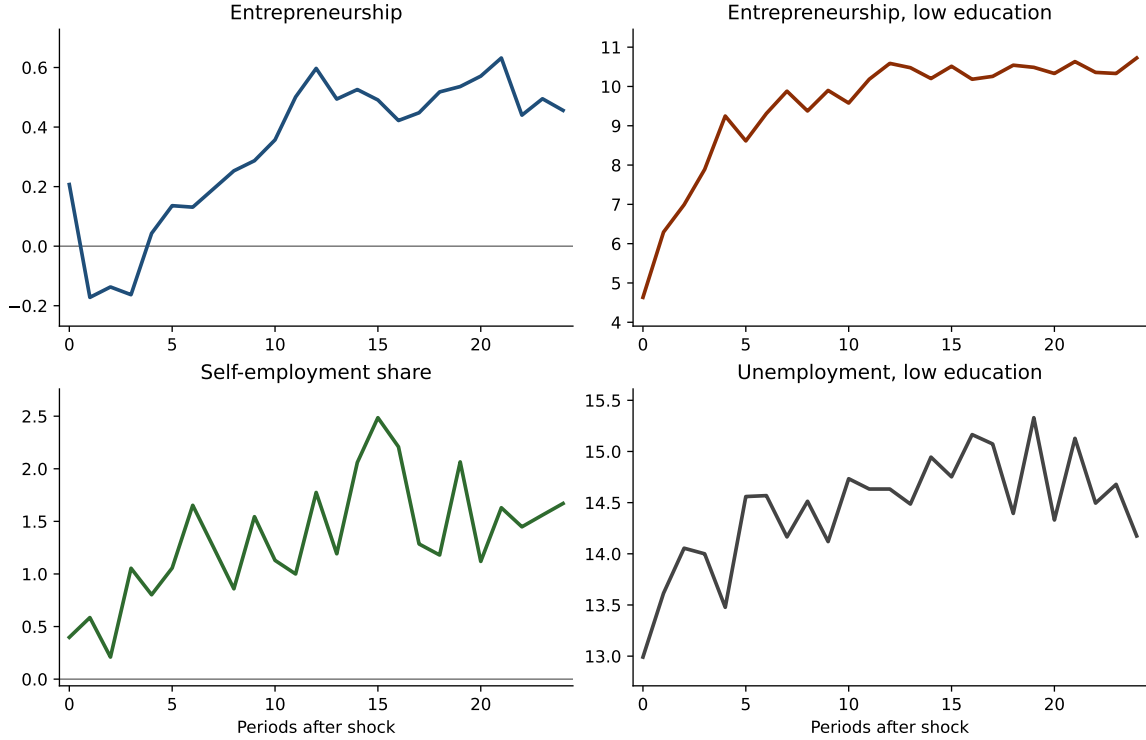
We model the downturn through the separation margin rather than through a simple MIT productivity shock, because the mechanism of interest operates through the outside option to entrepreneurship rather than through an exogenous fall in entrepreneurial productivity itself. A standard MIT productivity shock would move wages, firm profits, and entrepreneurial returns more broadly, making it difficult to isolate whether entrepreneurship changes because outside options deteriorate or because production opportunities themselves have changed. We compare the resulting steady state with the benchmark and then compute the deterministic transition from the benchmark distribution.

Table 5: Benchmark self-employment equilibrium: downturn steady state with higher low-education separation

	Benchmark	Downturn
Prices and taxes		
Interest rate r	0.097	0.097
Wage w	3.020	3.026
Job-finding rate θ	0.650	0.666
Income tax τ_y	0.007	0.007
Entrepreneur share		
All	0.178	0.187
Low-education	0.078	0.091
Mid-education	0.077	0.074
High-education	0.022	0.022
Self-employed share		
All	0.577	0.595
Low-education	0.364	0.410
Mid-education	0.199	0.172
High-education	0.015	0.013
Avg. entrepreneur size n		
All	6.159	5.798
Low-education	1.177	1.011
Mid-education	6.403	6.612
High-education	22.681	22.439
Output		
Agg. output index	1.000	0.979

Table 5 shows that the new steady state displays a modest increase in entrepreneurship, with the entrepreneur share rising from 0.178 to 0.187. At the same time, the self-employed share rises from 0.577 to 0.595, average entrepreneur size falls from 6.16 to 5.80 workers, and the output index declines from 1.000 to 0.979. The effect is concentrated among low-education individuals, for whom the entrepreneur share rises from 0.078 to 0.091. The recession shock therefore increases entrepreneurship only slightly in the aggregate, but shifts activity toward smaller-scale business ownership and away from more productive entrepreneurial scale.

Figure 6: Transition responses to a low-education separation shock.



Note: Panels report percentage-point deviations from the benchmark transition start state after a low-education separation shock. The top-left panel shows the overall entrepreneur share, the top-right the low-education entrepreneur share, the bottom-left the self-employment share, and the bottom-right the low-education unemployment share.

Figure 6 shows the transition dynamics behind this steady-state comparison. Following the shock, low-education unemployment rises on impact, and the entrepreneur pool shifts modestly toward self-employment. The figure sharpens the same interpretation as the steady-state comparison: a recession first worsens labor-market outcomes for exposed individuals and then pushes some marginal individuals into smaller-scale entrepreneurial activity. This is the necessity-entry channel emphasized in the paper.

9 Conclusion

This paper develops a quantitative framework for studying necessity entrepreneurship as an equilibrium outcome of occupational choice under labor-market frictions. By defining necessity and opportunity entrepreneurship structurally, rather than through stated motives or prior labor-market status, the paper provides a framework for analyzing how outside options shape entrepreneurial entry.

The paper's central message is that entrepreneurship is a compositional object, not just a scalar one. Labor-market conditions and policy affect not only the level of entrepreneurial entry,

but also whether that entry takes the form of lower-scale self-employment or employer-oriented entrepreneurship. This distinction matters for aggregate outcomes, because the same rise in business ownership can reflect either productive entrepreneurial opportunity or weak outside options.

The broader implication is that entrepreneurship should not be evaluated by aggregate entry alone. Understanding who becomes an entrepreneur, under what labor-market conditions, and at what scale is essential for interpreting both entrepreneurial dynamics and the effects of policy.

Appendix

Robustness

The employer-threshold benchmark is disciplined locally around $f_{hire} = 0.25$ rather than being treated as a single lucky parameter choice. The appendix reports nearby values to show that the main composition result is robust within a tight bracket.

Table 6: Employer-threshold cost robustness in the benchmark baseline equilibrium

	$f_{hire} = 0.15$	$f_{hire} = 0.25$	$f_{hire} = 0.35$
Entrepreneur share	0.1786	0.1781	0.1767
Self-employed share	0.5679	0.5768	0.5926
Employer share	0.4321	0.4232	0.4074
Avg. entrepreneur size n	6.1274	6.1587	6.2336
Avg. entrepreneur capital k	97.8587	98.1596	98.8487
Self-employed share by education			
Low-education	0.8190	0.8265	0.8494
Mid-education	0.4482	0.4577	0.4637
High-education	0.1109	0.1157	0.1215
Employer share by education			
Low-education	0.1810	0.1735	0.1506
Mid-education	0.5518	0.5423	0.5363
High-education	0.8891	0.8843	0.8785

Table 7: Employer-threshold cost robustness in the benchmark no-UI equilibrium

	$f_{hire} = 0.15$	$f_{hire} = 0.25$	$f_{hire} = 0.35$
Entrepreneur share	0.3348	0.3425	0.3401
Self-employed share	0.7321	0.7670	0.7710
Employer share	0.2679	0.2330	0.2290
Avg. entrepreneur size n	2.6566	2.5529	2.6138
Avg. entrepreneur capital k	55.6364	54.0490	54.7019
Self-employed share by education			
Low-education	0.8523	0.8711	0.8767
Mid-education	0.7324	0.7732	0.7767
High-education	0.4062	0.4408	0.4433
Employer share by education			
Low-education	0.1477	0.1289	0.1233
Mid-education	0.2676	0.2268	0.2233
High-education	0.5938	0.5592	0.5567

Varying f_{hire} between 0.15 and 0.35 barely changes the overall entrepreneur share in the baseline economy, but it moves composition in the expected direction: a higher employer threshold raises the self-employed share and lowers the employer share. The same qualitative pattern holds in the no-UI economy, where entrepreneurship remains high and increasingly concentrated in self-employment.

The matched fixed-tax UI ladder delivers the same qualitative comparative statics as the endogenous-tax benchmark. Lower UI still raises entrepreneurship and shifts the entrepreneur

pool toward smaller-scale self-employment when the financing rule is held fixed. The benchmark results therefore do not hinge on the fiscal closure used to finance unemployment insurance.

Measurement note

To compare aggregate production across steady states, let Y_s denote simulated total output in steady state s and let H denote the fixed number of simulated households. The aggregate output index reported in the quantitative tables is

$$\frac{Y_s/H}{Y_0/H} = \frac{Y_s}{Y_0},$$

where 0 denotes the benchmark steady state. So a value above one means output per household, and therefore total output, is higher than in the benchmark. Cutoff figures are omitted from the manuscript appendix for brevity.

Computation

Given the values for parameters, the discretized spaces Ω_a for a , Ω_z for z and Ω_x for x . Each agent draw a human capital level h (interpreted as educational attainment) at the beginning of the economy, which will be fixed henceforth. h will determine their labor productivity $z \in \Omega_z$, the separation rate ρ , and managerial ability $x \in \Omega_x$. Let $i_w = 0$ indicating being entrepreneur, $i_w = 1$ being employed and $i_w = 2$ being unemployed. The numerical algorithm works as follows.

1. Set a tolerance $\epsilon > 0$.
2. Guess $\Phi^0 = (r^0, w^0, \rho^0, \tau_y^0)$. Solve for optimal household behavior:

$$f : (\varphi; \Phi) \rightarrow (c, a', \mathcal{I}_e, n, k),$$

where $\varphi = \{h, z, x, a, i_w\}$. We will use the method of value function iteration as follows.

- (a) Guess value function $\mathbf{V}^0(\varphi; \Phi^0)$ and policy functions $f^0(\varphi; \Phi^0)$.
- (b) Calculate the cutoff values of x_u^* (for previously unemployed worker, or entrepreneur) and x_n^* (for previously employed worker) via the following conditions.

$$\tilde{w}_u = \pi(x_u^*) = \max_{k, n_h \geq 0} \left\{ x_u^* k^\alpha (\ell_e + \xi z n_h)^\gamma - w(1 + \psi\tau) z n_h - rk - \frac{\kappa \rho z n_h}{q(\theta)} - f_{hire} \mathbf{1}\{n_h > 0\} \right\} \quad (11)$$

$$\tilde{w}_n = \pi(x_n^*) = \max_{k, n_h \geq 0} \left\{ x_n^* k^\alpha (\ell_e + \xi z n_h)^\gamma - w(1 + \psi\tau) z n_h - rk - \frac{\kappa \rho z n_h}{q(\theta)} - f_{hire} \mathbf{1}\{n_h > 0\} \right\} \quad (12)$$

where $\tilde{w}_n$ and $\tilde{w}_u$ denote the expected compensation for the employed worker and unemployed worker respectively. More specifically,

$$\tilde{w}_n = (1 - \rho) \{zw(1 - (1 - \psi)\tau)\} + \rho b \quad (13)$$

$$\tilde{w}_u = p(\theta) \{zw(1 - (1 - \psi)\tau)\} p(\theta) b. \quad (14)$$

Noting that the separation rate ρ is pre-determined by the individual's initial human capital level h .

(c) Update value and policy functions:

$$\mathbf{V}^1(h, z, x, a, i_w) = \max_{\{a', c, \mathcal{I}_e\}} \left\{ \begin{array}{l} \nu_{i_w=0} \cdot \left[(1 - \mathcal{I}_e) \cdot \mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w)) \right. \\ \quad \left. + \mathcal{I}_e \cdot (V_e + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w = 2)) \right] + \\ \nu_{i_w=1} \cdot \left[(1 - \mathcal{I}_e) \cdot \mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w)) \right. \\ \quad \left. + \mathcal{I}_e \cdot (V_e + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w = 2)) \right] + \\ \nu_{i_w=2} \cdot \left[(1 - \mathcal{I}_e) \cdot \mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w)) \right. \\ \quad \left. + \mathcal{I}_e \cdot (V_e + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w = 2)) \right] \end{array} \right\} \quad (15)$$

$$f^1(\varphi; \Phi^0) = \arg \max \mathbf{V}^1(\varphi; \Phi^0)$$

where $\mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w))$ denotes the expected payoff for the individual who is previously unemployed worker, or entrepreneur, and decides to look for paid job, while $\mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w))$ represents the expected payoff for the previously employed worker who decide to stay in workplace.

$$\mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w)) = p(\theta) (V_n + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w = 1)) \quad (16)$$

$$+ (1 - p(\theta)) (V_u + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w = 0))$$

$$\mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w)) = (1 - \rho) (V_n + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w = 1)) \quad (17)$$

$$+ \rho (V_u + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w = 0))$$

To simplify the analysis, we assume that the individual who was previously employed and newly separated from the firm is not allowed to search for job at the current period. The measure for the individuals who are looking for jobs, u_t , the total vacancies v_t ,

and job matching rate $\theta_{n,t}$ are given as follows.

$$u_t = \left(\int_x \iota_{i_w=0} + \int_x \iota_{i_w=2} \right) \cdot \int_0^{x_u^*} \quad (18)$$

$$v_t = \int_x \iota_{i_w=1} \cdot \rho \quad (19)$$

$$\theta_{n,t} = \bar{m} \left(\frac{v_t}{u_t} \right)^{1-\mu} \quad (20)$$

Note that we can write the above value functions explicitly as a system of sub-problems: $P_{x < x_u^*}^{i_w=0}$, $P_{x \geq x_u^*}^{i_w=0}$, $P_{x < x_n^*}^{i_w=1}$, $P_{x \geq x_n^*}^{i_w=1}$, $P_{x < x_u^*}^{i_w=2}$, $P_{x \geq x_u^*}^{i_w=2}$, which correspond to the dynamic programming problem for i). previously self-employed with $x < x_u^*$; ii). previously self-employed with $x \geq x_u^*$; iii). previously employed with $x < x_n^*$; iv). previously employed with $x \geq x_n^*$; v). previously unemployed with $x < x_u^*$; vi). previously unemployed with $x \geq x_u^*$. To economize the space, we only provide the details for $P_{x < x_u^*}^{i_w=0}$, and $P_{x \geq x_u^*}^{i_w=0}$ as follows.

$$\left(P_{x < x_u^*}^{i_w=0} \right):$$

$$\mathbf{V}^1(h, z, x < x_u^*, a, i_w = 0) = \max_{a'_n, a'_u} \left\{ \begin{array}{l} p(\theta) (u(c_n) + \beta \mathbf{E} \mathbf{V}^0(h, z', x', a'_n, i'_w = 1)) \\ + (1 - p(\theta)) (u(c_u) + \beta \mathbf{E} \mathbf{V}^0(h, z', x', a'_u, i'_w = 0)) \end{array} \right\} \quad (21)$$

s.t.

$$c_n + a'_n \leq (1 - \tilde{r} - \delta)a + zw(1 - (1 - \psi)\tau), \text{ if matched with a firm} \quad (22)$$

$$c_u + a'_u \leq (1 - \tilde{r} - \delta)a + b, \text{ if unemployed} \quad (23)$$

$$\left(P_{x \geq x_u^*}^{i_w=0} \right):$$

$$\mathbf{V}^1(h, z, x \geq x_u^*, a, i_w = 0) = \max_{a'_e, k, n_h \geq 0} \{ u(c_e) + \chi_{se} \mathbf{1}\{n_h = 0\} + \chi_{emp} \mathbf{1}\{n_h > 0\} + \beta [(1 - \rho_e(h)) \mathbf{E} \mathbf{V}^0(h, z', x', a'_e, i'_w = 0)] \} \quad (24)$$

s.t.

$$c_e + a'_e \leq (1 - \tilde{r} - \delta)a + \pi(k, n_h, x; \tilde{r}, w, \tau) \quad (25)$$

where

$$\pi(k, n_h, x; \tilde{r}, w, \tau) = xk^\alpha (\ell_e + \xi z n_h)^\gamma - w(1 + \psi\tau)z n_h - rk - \frac{\kappa \rho z n_h}{q(\theta)} - f_{hire} \mathbf{1}\{n_h > 0\}. \quad (26)$$

- (d) Stop if $\max \{ |\mathbf{V}^1 - \mathbf{V}^0|, |f^1 - f^0| \} \leq \epsilon$. Set $\mathbf{V}^* = \mathbf{V}^1$, and $f^* = f^1$. Otherwise, set $\mathbf{V}^0 = \mathbf{V}^1$, $f^0 = f^1$ and repeat from step (c).

3. Generate a large number of individuals, $N = 100000$. For each agent j assign a vector of

initial condition $(h_t^j, z_0^j, x_0^j, a_0^j, i_{w,0}^j)$, where $x_0^j \sim \Omega_x$, $z_0^j \in \Omega_z$, $i_{w,0}^j \in \Omega_w$.

4. Simulate the economy for T periods, where T is sufficiently large.

5. Calculate the following statistics from the simulated path $\{h_t^j, z_t^j, x_t^j, a_t^j, i_{w,t}^j, \mathcal{I}_{e,t}^j, n_t^j, k_t^j\}_{t=0}^T$.

$$LS^0 = \frac{\sum_{j=1}^N (\iota_{i_w=1} \cdot z^j)}{N} \quad (27)$$

$$KS^0 = \frac{\sum_{j=1}^N (\iota_{i_w=0} \cdot a^j + \iota_{i_w=1} \cdot a^j + \iota_{i_w=2} \cdot a^j - \iota_{i_w=0} \cdot k^j)}{\sum_{j=1}^N a^j} \quad (28)$$

$$u_t = \left(\int_h \iota_{i_w=0} + \int_h \iota_{i_w=2} \right) \cdot \int_0^{h_u^*} \quad (29)$$

$$v_t = \int_h \iota_{i_w=1} \cdot \rho \quad (30)$$

$$p(\theta)_t = \bar{m} \left(\frac{v_t}{u_t} \right)^{1-\mu} \quad (31)$$

$$G = \int \tau w z \quad (32)$$

and τ_y^1 that balances the government's budget.

6. Stop and set $(r^*, w^*, \theta^*) = (r^0, w^0, \theta^0)$, if $\max \{LS^0, KS^0, |\theta_n^1 - \theta_n^0|\} \leq \epsilon$. Otherwise, update aggregate variables (restart from step 2):

$$\begin{aligned} r^0 &= \chi r^0 + (1 - \chi) r^1 (KS^0) \\ w^0 &= \chi w^0 + (1 - \chi) w^1 (LS^0) \\ p(\theta)^0 &= \chi p(\theta)^0 + (1 - \chi) p(\theta)^1 \\ \tau_y^0 &= \chi \tau_y^0 + (1 - \chi) \tau_y^1 \end{aligned}$$

where $\chi \in (0, 1)$ is the step for updating aggregate variables.

Appendix Search and Matching

I am not sure how much of this is needed for coding purposes but this is the standard search and matching model. The stock of employed workers n_t can be expressed as the current pool of workers who did not separate from firms, plus new hires

$$n_t = (1 - \rho)n_{t-1} + m_{t-1}$$

The matching function is Cobb-Douglass where

$$m_t = \bar{m} v_t^{1-\mu} u_t^\mu$$

$\bar{m}$ is a match efficiency parameter, v is vacancies, u is unemployment, and match elasticity is $0 < \mu < 1$. The hiring rate can be defined for firms as total number of new hires to the number of vacancies

$$q(\theta_t) = \frac{m_t}{v_t} = \bar{m} \theta_t^{-\mu}$$

where $\theta_t = \frac{v_t}{u_t}$ is aggregate labour market tightness. Similarly we can define this from the perspective of the worker in the job-finding rate.

$$p(\theta_t) = \frac{m_t}{u_t} = \bar{m} \theta_t^{1-\mu}$$

The Job Creation Condition determines the amount of new vacancies an employer will post. In our model, most workers are on rolling contracts, so firms only need to post new vacancies after some workers separate from these rolling contracts. Usually in search and matching models the wage is determined ex-post of matching by Nash bargaining between the employer and employee. However, in our span-of-control model the wage has already been determined ex ante, as the wage is an endogenous variable that determines occupational choice. The marginal benefit that the new employee would bring to the firm is expressed as A (which can be derived from the firms production function, $\gamma X k^\alpha (zn)^{\gamma-1}$). The *JCC* can be then expressed as the costs of replacing out workers on the *LHS* and the expected benefit on the *RHS*

$$\frac{\kappa}{q(\theta_t)} = A - w_t + \beta E_t \left[A - w_{t+1} + (1 - \rho) \frac{\kappa}{q(\theta_{t+1})} \right]$$

where κ is the cost of posting a vacancy.

Matching Steady States

To find the steady state of the employment equation $n_t = (1 - \rho)n_{t-1} + m_{t-1}$, we use the fact that $u = (1 - n)$ and $m_t = \bar{m} v_t^{1-\mu} u_t^\mu$. Substituting and getting rid of time subscript gives $(1 - u) = (1 - \rho)(1 - u) + \bar{m} v^{1-\mu} u^\mu$ which simplifies to $\bar{m} v^{1-\mu} u^\mu = \rho(1 - u)$. Making v the subject of the formula gives us the Beveridge curve.

$$v = \left(\frac{\rho(1 - u)}{\bar{m} u} \right)^{\frac{1}{1-\mu}}$$

The steady state for the JCC can be found by using $\frac{\kappa}{q(\theta_t)} = A - w_t + \beta E_t \left[A - w_{t+1} + (1 - \rho) \frac{\kappa}{q(\theta_{t+1})} \right]$ and $q(\theta_t) = \bar{m} \theta_t^{-\mu} = \bar{m} \left(\frac{v_t}{u_t} \right)^{-\mu}$. Substituting and getting rid of

time subscripts yields

$$\frac{\kappa}{\bar{m}} \left(\frac{v}{u} \right)^\mu = A - w_t + \beta \left[A - w + (1 - \rho) \frac{\kappa}{\bar{m}} \left(\frac{v}{u} \right)^\mu \right]$$

Simplifying yields the steady state of the JCC

$$1 - \beta(1 - \rho) \frac{\kappa}{\bar{m}} \left(\frac{v}{u} \right)^\mu = (1 + \beta) [A - w]$$

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